2. Calculate voltage and current

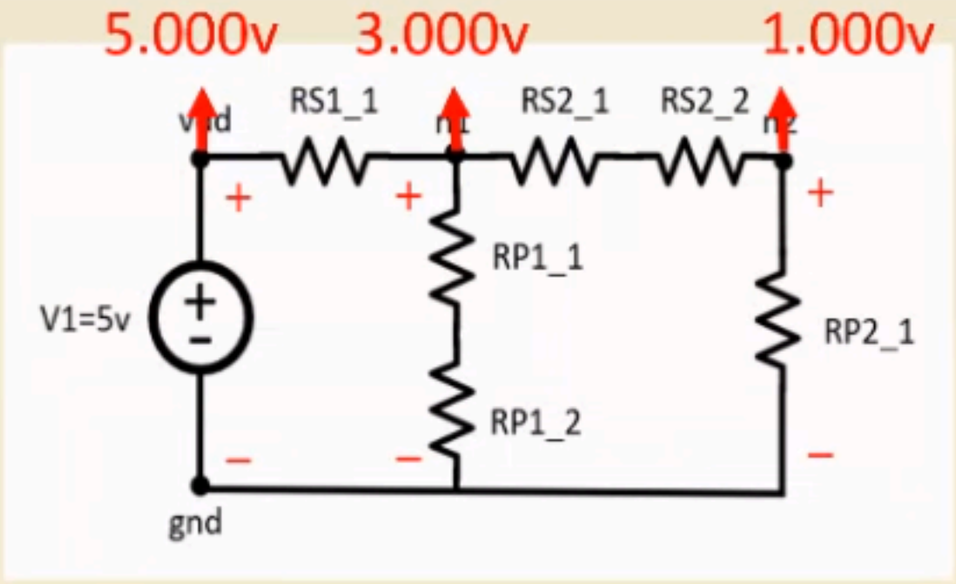
- With KCL and KVL, we can easily calculate **current cross each resistor** and **voltage of each node**.
- The execution should show:

```
==========================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 2
==========================================
Calculating...
The result is successfully calculated!
```

2. Calculate voltage and current

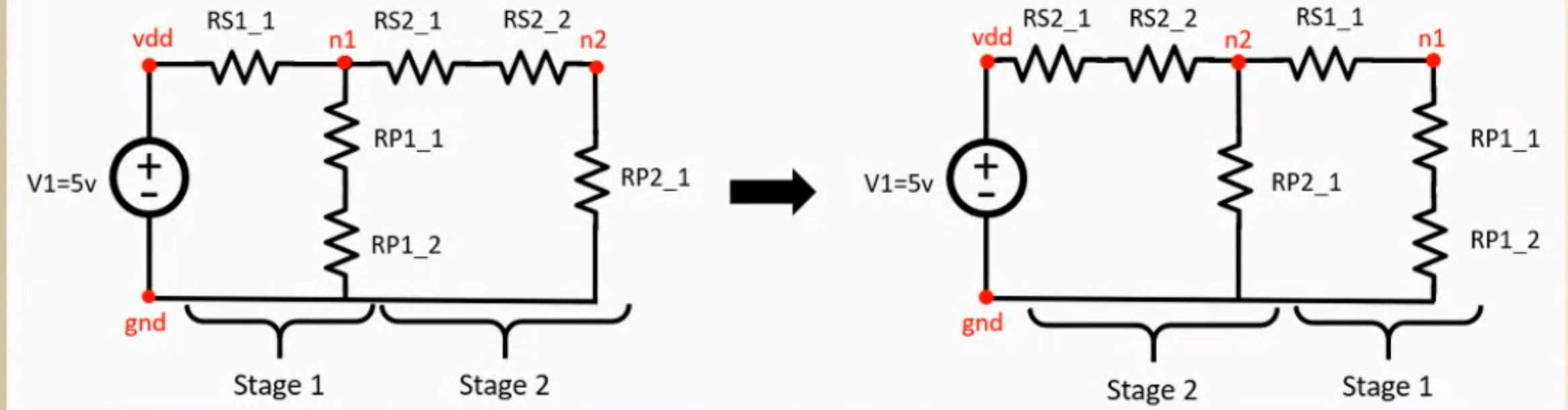
◆The calculation results(voltage of each node):

```
**********************************************
**                <<voltage>>                 **
**      vdd              5.000v              **
**      n1               3.000v              **
**      n2               1.000v              **
**                <<current>>                 **
**      RS1_1            1.000mA             **
**      RP1_1            500.000uA           **
**      RP1_2            500.000uA           **
**      RS2_1            500.000uA           **
**      RS2_2            500.000uA           **
**      RP2_1            500.000uA           **
**                                           **
**                       myspice2021         **
**********************************************
```



3. swap stage

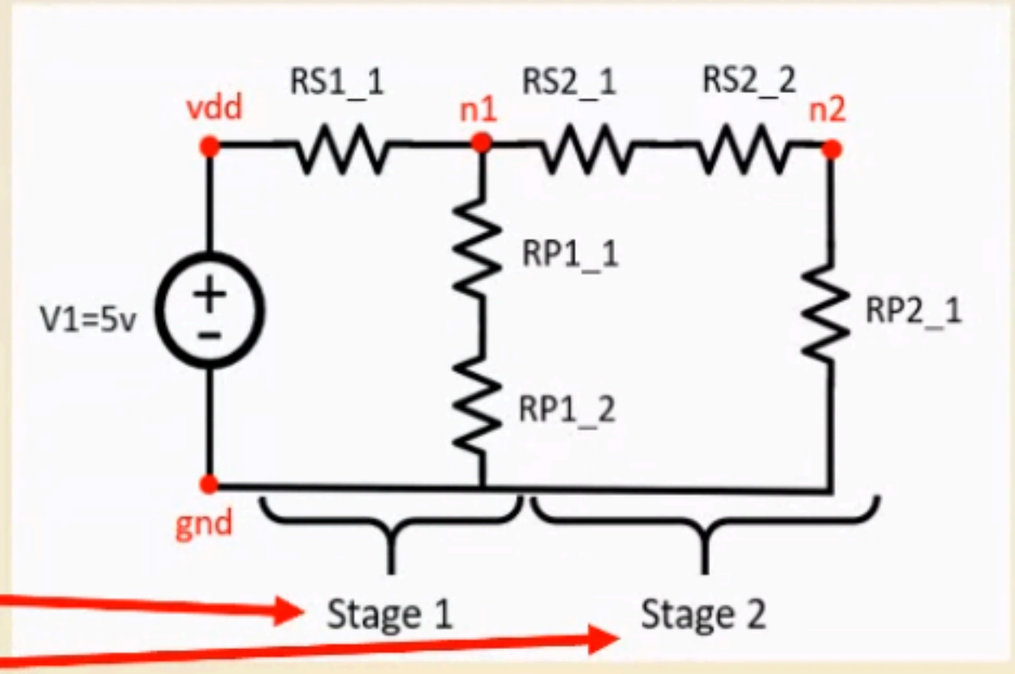
- The program will swap two stages the user entered.



3. swap stage

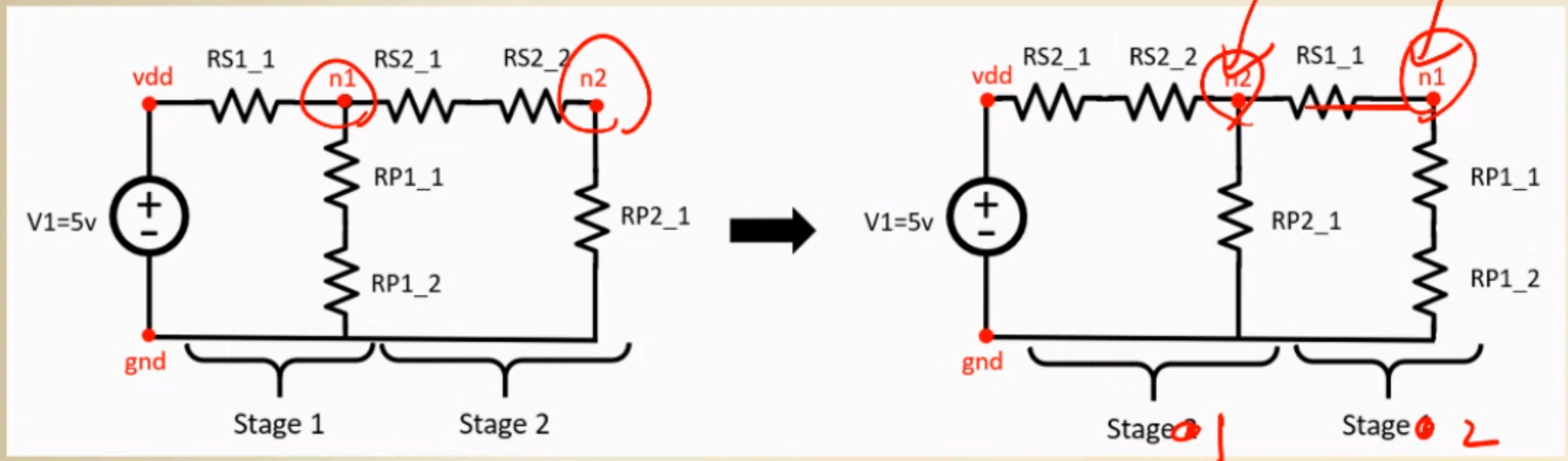
◆The execution should show:

```
==========================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 3
==========================================
Please enter the first stage you want to swap: 4
The node doesn't exist. Please enter again.
Please enter the first stage you want to swap: 1
Please enter the second stage you want to swap: 2
RS1_1 is changing node from vdd to n2 ...
RS2_1 is changing node from n1 to vdd ...
The stage is Successfully swapped!
```



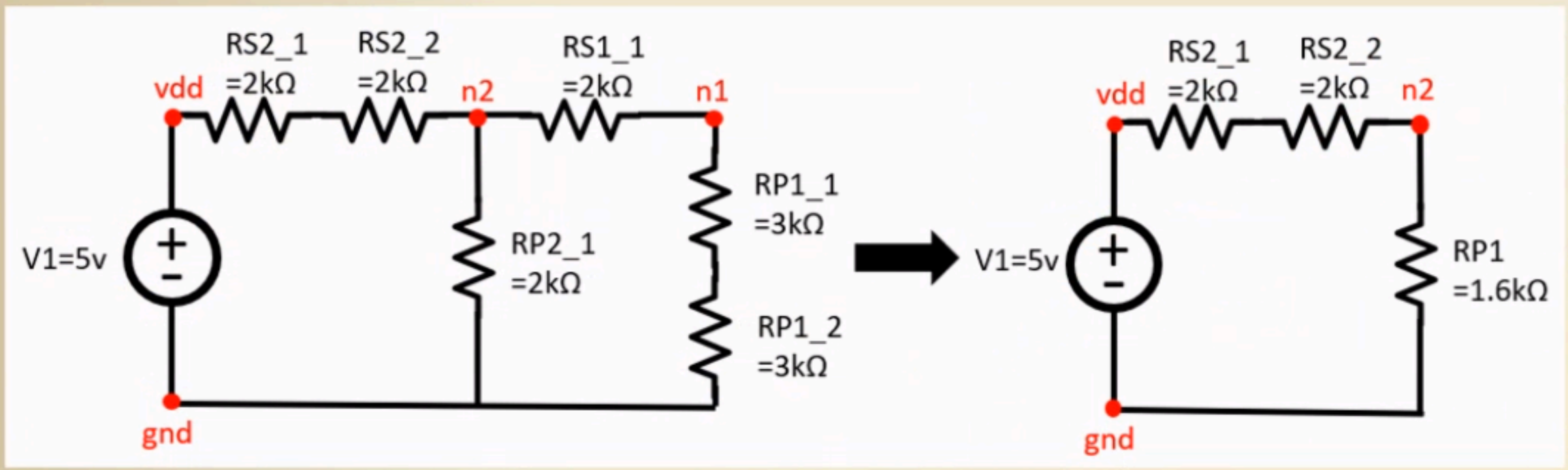
3. swap stage

- The program will swap two stages the user entered.



4. merge stage

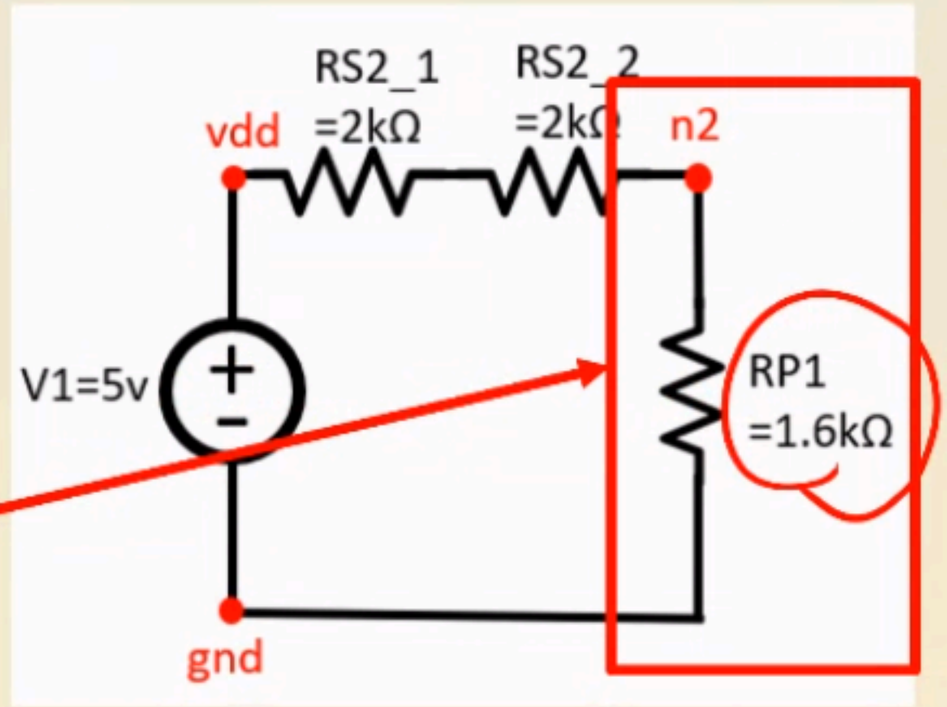
- The program will merge two stages, which the user entered, into first stage.



4. merge stage

◆The execution should show:

```
==========================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 4
==========================================
Please enter the first stage you want to merge: 1
Please enter the second stage you want to merge: 2
Resistors are merged into RP1.
=> RP1 n2 gnd 1.600 k
The stage is Successfully merged!
```







5. Output file

- If user want to extract the calculated result, the program will output the **result.txt** file. The output file should contain the result of the calculations and the circuit descriptions.
- The execution should show:

```
==========================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 5
==========================================
Please enter the name of the output file: result.txt
Exporting the file...
The output file successfully exported!
```



5. Output file

Output file

```
***circuits***
stage 1

v1 vdd gnd 5V

RS2_1 vdd ns2_1 2k
RS2_1 ns2_1 n2 2k
RP1 n2 gnd 1.600k
```

```
***calculation results***
**********************************************
**                    <<voltage>>             **
**        vdd         5.000v                  **
**        n1          3.000v                  **
**        n2          1.000v                  **
**                    <<current>>             **
**        RS1_1       1.000mA                 **
**        RP1_1       500.000uA               **
**        RP1_2       500.000uA               **
**        RS2_1       500.000uA               **
**        RS2_2       500.000uA               **
**        RP2_1       500.000uA               **
**                                            **
**                                  myspice2021 **
**********************************************
```

the circuit descriptions (Program should print the circuit before user press 5 in the main menu)

the result of the calculations (Program should print the results, which is from the last time when user press 2 in the main menu)

5. Output file

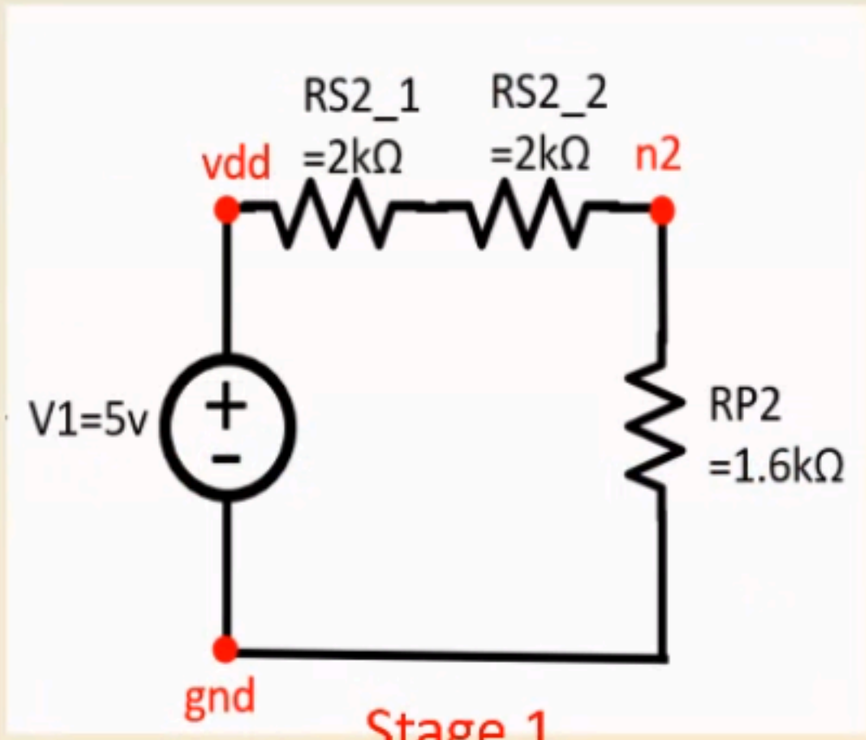
Output file

circuits

stage 1

v1 vdd gnd 5V

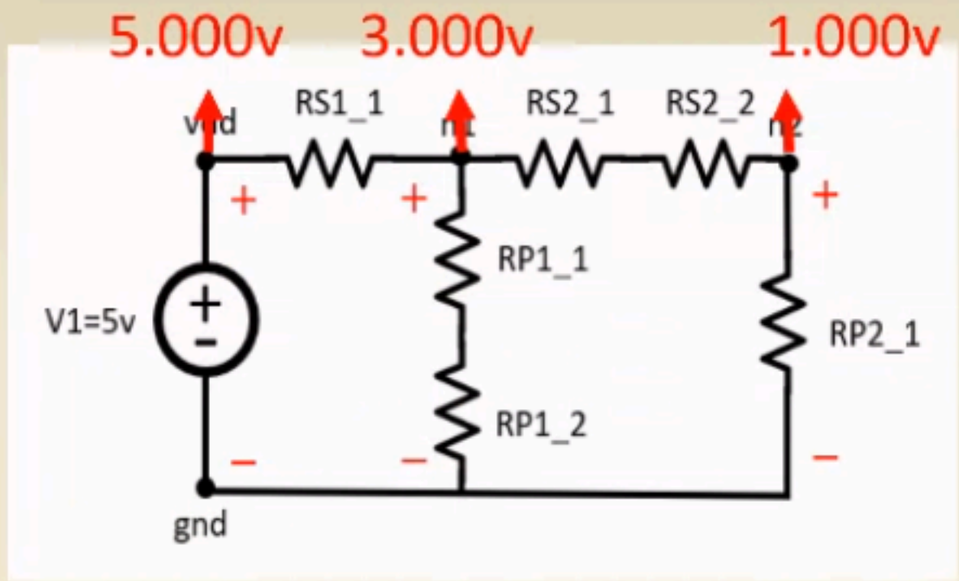
RS2_1 vdd ns2_1 2k
RS2_1 ns2_1 n2 2k
RP1 n2 gnd 1.600k

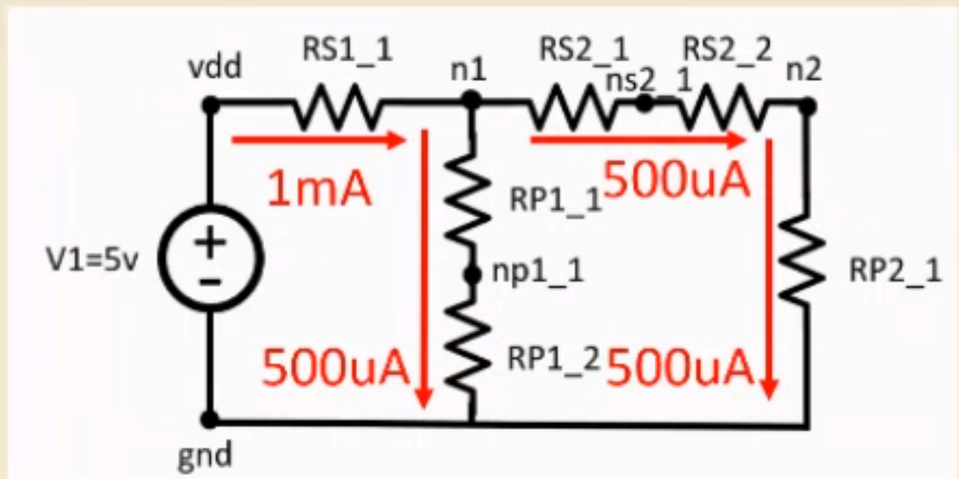


5. Output file

Output file

```
***calculation results***
**********************************************************
**                    <<voltage>>              **
**        vdd         5.000v                   **
**        n1          3.000v                   **
**        n2          1.000v                   **
**                    <<current>>              **
**        RS1_1       1.000mA                  **
**        RP1_1       500.000uA                **
**        RP1_2       500.000uA                **
**        RS2_1       500.000uA                **
**        RS2_2       500.000uA                **
**        RP2_1       500.000uA                **
**                                             **
**                    myspice2021              **
**********************************************************
```









6. Exit

- Enter 6 to exit.
- The execution should show:

```
==========================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 6
==========================================
Bye~
```



1. Input file

- An example of input file:

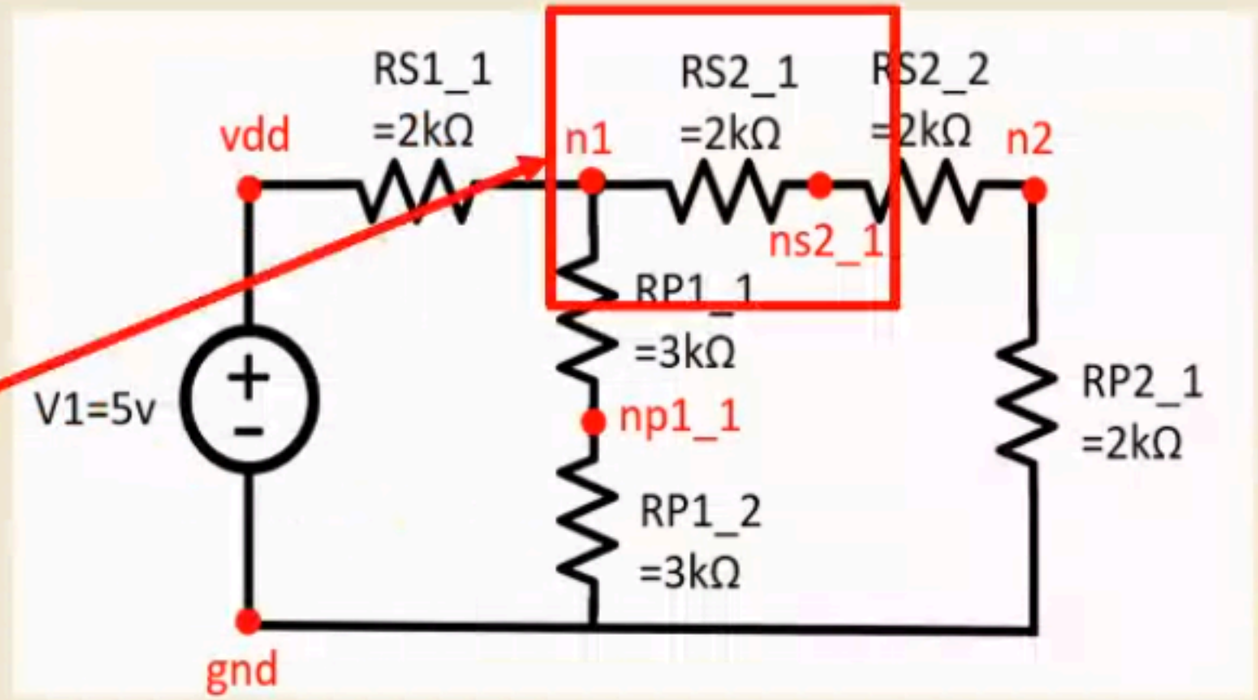
Input file

```
stage 2

v1 vdd gnd 5V

RS1_1  vdd     n1      2K
RP1_1  n1      np1_1   3K
RP1_2  np1_1   gnd     3K
RS2_1  n1      ns2_1   2K
RS2_2  ns2_1   n2      2K
RP2_1  n2      gnd     2K

end
```



1. Input file

◆How to get input information:

```
void input_file(int stages, int voltage, int resistor[15][], string node_name[][]){
    char data[];

//input file
    ifstream infile();
    seekg();
    infile >> data;
    stages = atoi(data[]);

//voltage source
    seekg()
    infile >> data;
    voltage = atoi(data[]);

//initialize resistor & node name
    seekg();
    while(){
        infile >> data;
        resistor[][] = ;
        node_name[][] = ;
        resistor_num = ;

    }
}
```

2. calculate voltage and current

◆思考方向:

先得出最簡化的電路，只有一個電壓源跟一個等效電阻，就可以求出總電流了

→

算出對每個node往後看得等效電阻可以簡化問題

→

利用分流分壓公式、算出壓降等等方式

→

得出電流電壓

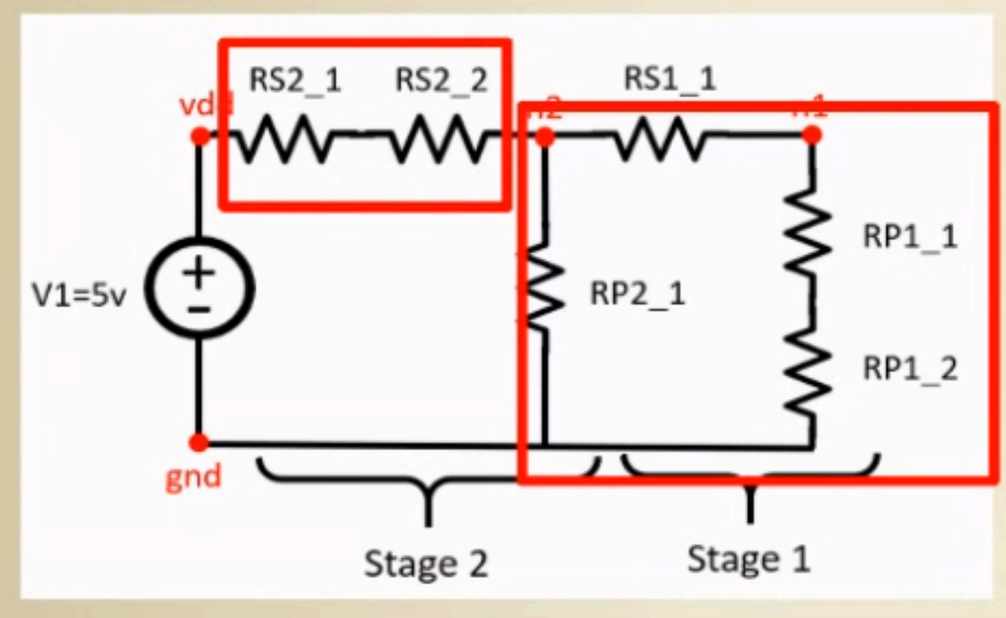
2. calculate voltage and current

◆思考方向:

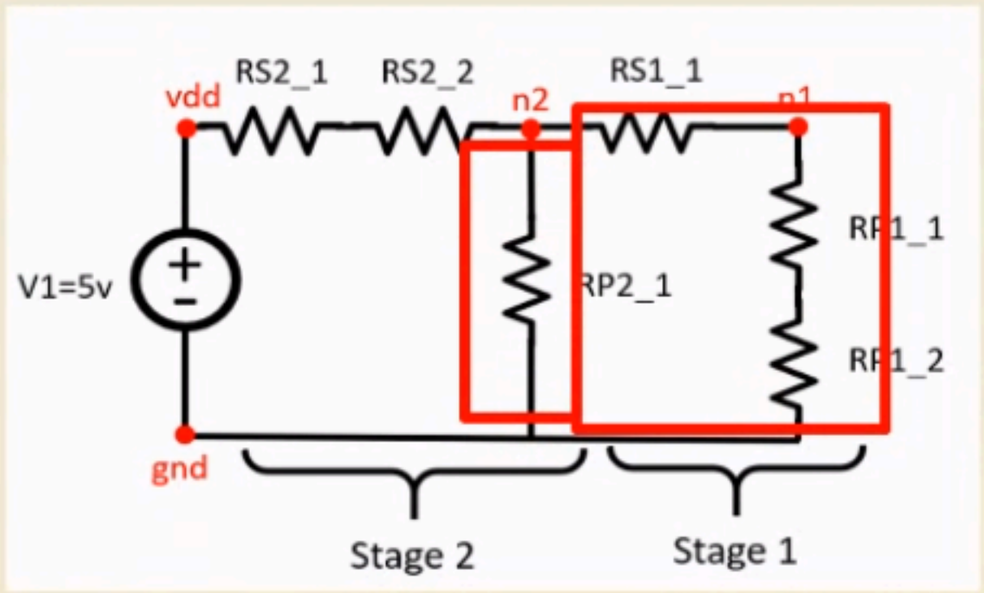
先得出最簡化的電路，只有一個電壓源跟一個等效電阻，就可以求出總電流了 → 算出對每個node往後看得等效電阻可以簡化問題 → 利用分流分壓公式、算出壓降等等方式 → 得出電流電壓

2. calculate voltage and current

◆思考方向:



看成串聯兩個等效電阻
可以算分壓



看成並聯兩個等效電阻
可以算分流

2. calculate voltage and current

◆How to get calculate voltage and current :

```
void calculate(int stages, int voltage, int resistors[], int resistor_num[], int V_node[], int I_resistor[][]){
    int total_i;
    int eq_resistor[15][2];

//the equivalent resistor
    for(int i=0; i< stages; i++){
        eq_resistor[][] = ;
    }

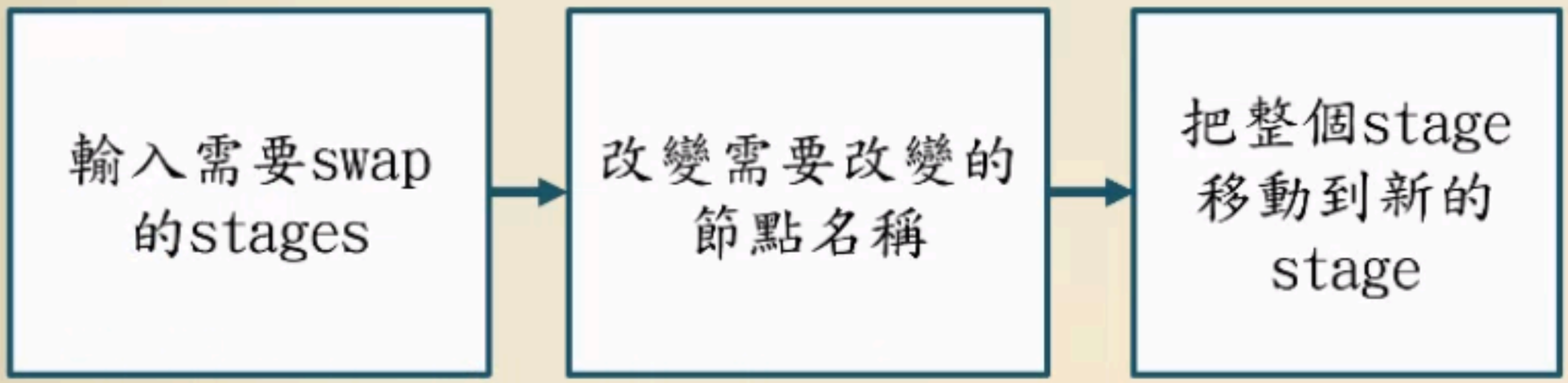
//total current
    total_i = voltage / eq_resistor[][];

//calculation voltage & current
    for(int i=0; i< stages; i++){
            for(){
                v_node[] = voltage - I_resistor * eq_resistor[][];
                I_resistor[][] = v_node[] / eq_resistor[][];
            }
    }

//print results
    for(int i=0; i< stages; i++){
        cout << left << setw() << "**";
        cout << right << setw() << "**";
    }
}
```

3. swap stage

◆思考方向:



3. swap stage

◆How to swap stage :

```
void swap_stages(int resistors[15][], int resistor_num[], string node_name[][]){
// input

//swap
    node_nume[][] = ; //change the name of the node
    node_nume[][] = ;
    if(stage is not at the end of the circuit){
        node_nume[][] = ;
    }

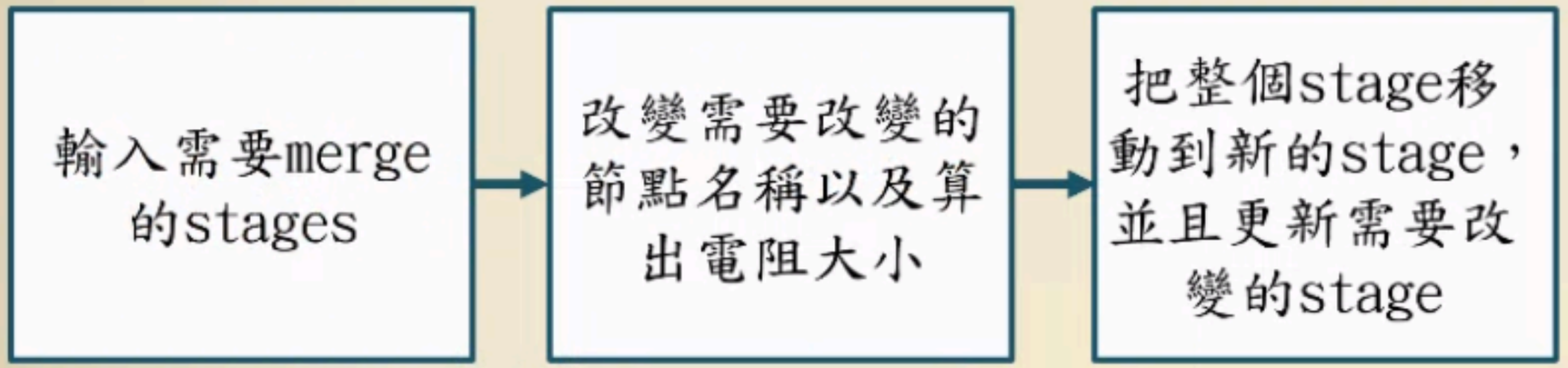
    for(){ //change the stage of the node
        node_name[][] = ;
        resistors[][] = ;
        resistor_num[] = ;
    }


//cout

}
```

4. merge stage

◆思考方向:



4. merge stage

◆How to get merge stage :

```
void merge_stages(int resistors[15][], int resistor_num[], string node_name[][){
// input

//swap
    node_nume[][] = ; //change the name of the node
    resistors[][] = ;
    if(stage is not at the end of the circuit){
        node_nume[][] = ;
    }

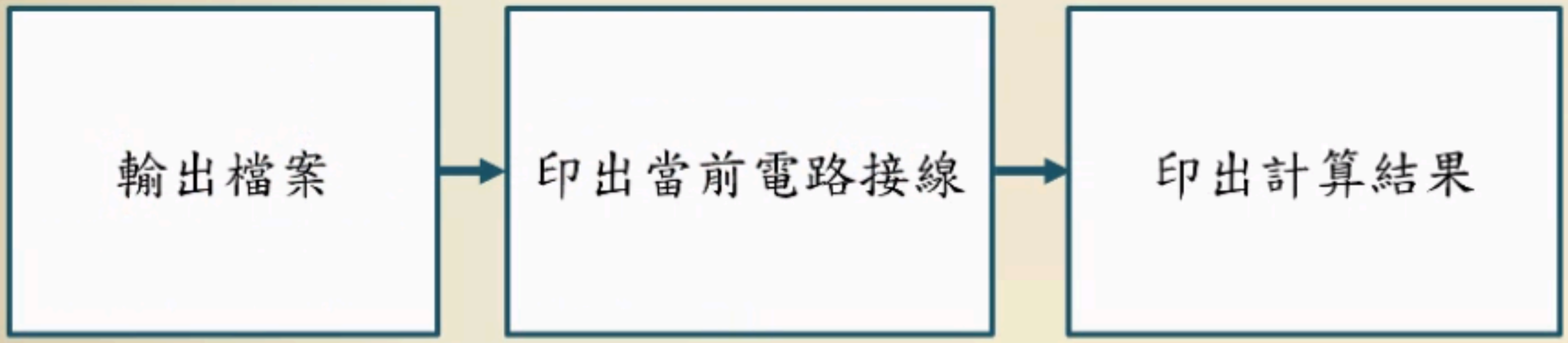
    for(){ //change the stage of the node
        node_name[][] = ;
        resistors[][] = ;
        resistor_num[] = ;
    }


//cout

}
```

5. Output file

◆思考方向:



5. Output file

◆How to get input information:

```
void output_file(int stages, int voltage, int resistors[15][], int resistor_num[],
                 string node_name[][], int V_node[], int I_resistor[][]){

//output file
    ofstream outfile();


//circuit description
    for(){
        outfile << resistors[][] << node_name[][];
    }


//calculation results
    for(){
        outfile << node_name[][] << V_node[];
    }
    for(){
        outfile << resistors[][] << I_resistor[][];
    }
}
```

5. Output file

◆How to get input information:

```
void output_file(int stages, int voltage, int resistors[15][], int resistor_num[],
                 string node_name[][], int V_node[], int I_resistor[][]){

//output file
    ofstream outfile();


//circuit description
    for(){
        outfile << resistors[][] << node_name[][];
    }


//calculation results
    for(){
        outfile << node_name[][] << V_node[];
    }
    for(){
        outfile << resistors[][] << I_resistor[][];
    }
}
```

5. Output file

```
***calculation results***
**********************************************
**                <<voltage>>                 **
**   vdd              5.000v                  **
**   n1               3.000v                  **
**   n2               1.000v                  **
**                <<current>>                 **
**   RS1_1            1.000mA                 **
**   RP1_1          500.000uA                 **
**   RP1_2          500.000uA                 **
**   RS2_1          500.000uA                 **
**   RS2_2          500.000uA                 **
**   RP2_1          500.000uA                 **
**                                            **
**                myspice2021                 **
**********************************************
```

4 spaces

12 spaces

16 spaces

36 spaces

20 spaces

Figure. 5 The format of the calculation result.

5. Output file

```
***calculation results***
**********************************************
**                <<voltage>>                  **
**   vdd              5.000v                   **
**   n1               3.000v                   **
**   n2               1.000v                   **
**                <<current>>                  **
**   RS1_1            1.000mA                  **
**   RP1_1          500.000uA                  **
**   RP1_2          500.000uA                  **
**   RS2_1          500.000uA                  **
**   RS2_2          500.000uA                  **
**   RP2_1          500.000uA                  **
**                                             **
**                myspice2021                  **
**********************************************
```

4 spaces

12 spaces

16 spaces

36 spaces

20 spaces

Figure. 5 The format of the calculation result.





8.6 Updating Sequential Files

- Another approach is fixing the field width in advance to leave enough space for future updating.
- For example, we can set the maximum length of *name* is 11
  - `outClientFile << account << ' ' << setw(11) << name << ' ' << balance <<endl;`
  - Original File: `300 White 0.00`
  - Updated File: `300 Worthington 0.00`
  - Since the total length of this record (including space) is not changed, this update will not affect the following records.
- Although this approach does not damage other data in the same file, more redundant spaces will make the file larger.
  - Length checking is also required on the input data.



```cpp
// Fig. 22.25: fig22_25.cpp
// Using strtok to tokenize a string.
#include <iostream>
#include <cstring> // prototype for strtok
using namespace std;

int main()
{
   char sentence[] = "This is a sentence with 7 tokens";
   char *tokenPtr;

   cout << "The string to be tokenized is:\n" << sentence
      << "\n\nThe tokens are:\n\n";

   // begin tokenization of sentence
   tokenPtr = strtok( sentence, " " );

   // continue tokenizing sentence until tokenPtr becomes NULL
   while ( tokenPtr != NULL )
   {
      cout << tokenPtr << '\n';
      tokenPtr = strtok( NULL, " " ); // get next token
   } // end while

```